FED4FED: Statistical Fairness Evaluation Report

Privacy-Preserving Verification for Federated Learning Models

FAIRNESS VERDICT: FAIR (Performance Homogeneous)

Under the Conservative Joint Policy (Disparity detected if either Alg 1 or Alg 2 rejects H0), the global federated model demonstrates statistically fair performance. Neither test found significant evidence to reject the null hypothesis of equal performance at alpha = 0.05. Alg 1 p-value: 0.3883, Alg 2 p-value: 0.4010.

1. Experiment & Architecture Configuration

Experiment ID:	cifar-10_fair_20261013_001339	**Date / Time:**	2026-10-13 00:14:50
Dataset:	CIFAR-10	**Scenario:**	Fair
Clients (K):	25	**Seed:**	42
FL Rounds:	5	**Final Val Accuracy:**	0.2951
Partition Method:	uniform	**Bootstrap Replications (B):**	200

2. Statistical Fairness Hypothesis Tests

Test Algorithm	Statistic	df	p-value	Significance (α)	Decision
Algorithm 1 (Graybill-Deal / T1)	25.3239	24	0.3883	0.050	**Fail to Reject (Fair)**
Algorithm 2 (Welch-Type F)	1.0462	(24, 1796.5)	0.4010	0.050	**Fail to Reject (Fair)**

3. Algorithm 3: Global Performance Confidence Interval

Status: Computed under homogeneous fairness condition.
Confidence Level: 95.0%
Pooled Performance Estimate (Graybill-Deal): 0.2945
Confidence Interval: [0.2857, 0.3034]
Standard Error: 0.0045 (Effective df = 1796.5)

4. Sample Decentralized Client Performance Statistics

Client ID	Test Samples (n_i)	Observed Metric	Bootstrap Mean	Variance (S_i^2)	Std Error (SE)
Client 00	400	0.3575	0.3594	0.000527	0.0230
Client 01	400	0.2800	0.2794	0.000401	0.0200

Client 02	400	0.3050	0.3055	0.000453	0.0213
Client 03	400	0.2850	0.2877	0.000553	0.0235
Client 04	400	0.2450	0.2441	0.000451	0.0212
Client 05	400	0.3175	0.3172	0.000570	0.0239
Client 06	400	0.2750	0.2754	0.000592	0.0243
Client 07	400	0.2875	0.2884	0.000489	0.0221

Scientific Methodology: Fed4Fed privacy-preserving statistical homogeneity testing using non-parametric bootstrapping, Graybill-Deal common mean estimation, and Welch-type heteroscedastic F-testing. Generated by FED4FED Research Suite v1.0.0.